

Agile! The Good, the Hype, and the Ugly (Bertrand Meyer)

Book Summary

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This is a practical book that enables the reader to benefit from the good ideas of agile methods while staying away from the bad ones. Agile methods are a mix of horrible and great ideas. In order to spare the reader from having to try out Scrum, Extreme Programming, Lean Software, and Crystal all on his own, this book provides a description and assessment of their underlying ideas.

First, those methods are described without the usual sermons and anecdotes. Second, the underlying ideas are scrutinized and assessed, thereby separating the useful from the harmful ones. Agile methods and their texts put three difficulties in the reader's way:

1. *Partisanship*: While it's common for their inventors to argue in favour of their inventions, proponents of agile methods ignore the rules of rational discourse and ask the reader to join their cult, which is inappropriate for engineering problems.
2. *Intimidation*: Agile texts dismiss previous approaches to software development as out-dated and their users as rigid and backwards. Who wants to argue against "agile" when all of this word's opposites have negative meanings?
3. *Extremism*: Proponents of some agile methods insist on the application of a method in its entirety, which discourages a more differentiated view on the techniques that make up a specific method.

While previous books criticizing agile methods were rather timid, this book will call out the bad ideas without undue dereference.

The first chapter is a summary of agile ideas. In the second chapter, the rhetoric of agile texts is analyzed. Chapter three is a description of traditional plan-based software development—to which agile methods are the antidote. Chapters four to eight review agile ideas: principles, roles, practices, and artifacts.

Those chapters do not focus on individual methods, but on their many commonalities. Scrum, Lean, XP, and Crystal are then presented in chapter nine as combinations of those principles already discussed, each with their one single big idea emphasized.

Chapter ten describes precautions organizations should take when adopting agile methods. Chapter eleven is the final assessment; it classifies agile ideas into three categories: the good, the hype, and the ugly.

This book is not a comprehensive guide to software development, but a critique of existing approaches. Authors usually argument by gut feeling, by experience (e.g. projects saved by agile approaches and ruined by ignoring them), logical reasoning, and, ideally, empirical analysis (for which there is usually too little data available). This book relies mostly on analytical reasoning, combined with personal experience and anecdotes for illustrative purposes.

1 Overview

While agile ideas date back to the 1990s, the movement gained traction with the publication of the *Manifesto for Agile Software Development* in 2001, which was signed by many proponents of existing agile ideas. Agile software development is not a single method, but a collection of ideas grouped together to methodologies such as *Extreme Programming*, *Lean Software Development*, *Crystal*, and *Scrum*, which all select and prioritize their own subset of agile ideas. However, there are some common core characteristics to those methods:

- *Values*: general assumptions framing the agile world view
- *Principles*: organizational and technical rules based on those values
- *Roles*: responsibilities and privileges of actors involved
- *Practices*: specific activities based on said principles

- *Artifacts*: virtual and physical tools supporting those practices

1.1 Values

The fundamental assumptions of agile software development are captured in the following values:

1. *Redefined roles for developers, managers, and customers*: Some of the manager's duties are transferred to the team, such as the selection and assignment of tasks. The developers and their code are moved into the center. Customers are no longer passive recipients of a product but active participants in the development process.
2. *No "big upfront" steps*: Activities preceding the writing of code such as gathering requirements ("The customer does not know what he wants!") or creating a design ("The developers do not know what will work!") are left out because they are subject to change anyway. Instead, requirements and design emerge in a continuous process involving the customer, in which the software is iteratively refactored into his acceptance.
3. *Iterative development*: Development takes place in iterations of a fixed time, usually a few weeks, for which the functionality with the highest business value is implemented by working through a prioritized list of tasks. Functionality is added iteratively.
4. *Limited, negotiated functionality*: Only the most important features, measured by their business value, will be implemented. Unused functionality is deemed wasteful and therefore not implemented in the first place. The functionality to be added is negotiated before the start of every iteration. Since it is empirically impossible to fix both scope and deadline, usually the deadline is retained, but the scope limited accordingly.
5. *Focus on quality, understood as achieved through testing*: Quality is ensured by continuous testing rather than by upfront design decisions or by sticking to development methodologies. The project's regression test suite is a central artifact and must always pass when new functionality is added.

1.2 Principles

The following principles—not the ones from the original Manifesto, but extracted from the various texts on agile software development—turn said values into prescriptions:

- *Organizational*
 1. *Put the customer at the center*: Customer representatives are involved throughout the project, which should deliver the best Return on Investment to the customer.
 2. *Let the team self-organize*: The team members pick their own tasks rather than having them assigned by a manager.
 3. *Work at a sustainable pace*: Programmers work reasonable hours rather than through intense phases with long hours to meet deadlines set by a manager.

4. *Develop minimal software*: 1) Only essential functions are built (*minimal functionality*); 2) Only what is requested is built, and extra work for future reuse and extensibility is left out (*minimal product*); 3) Only what is delivered to the customer—programs and tests—is built (*minimal artifacts*).
5. *Accept change*: Requirements are not complete at the beginning, but evolve as customers interact with intermediate releases. Change is considered normal.

- Technical

1. *Develop iteratively*: Every iteration of a fixed number of weeks produces a working release, providing the customer new functionality he can try out and give feedback on, which is then considered for a later iteration. No new functionality is demanded during an ongoing iteration; such requests are taken into consideration for an upcoming iteration instead.
2. *Treat tests as a key resource*: Quality is ensured by automated tests. No new development must start until all current tests pass. A text expresses the requirements new code must satisfy and is therefore written before the production code (*test first*).
3. *Express requirements through scenarios*: A scenario (e.g. a *use case* or a *user story*) describes an interaction of a user with the system that is to be built. Scenarios are obtained from the customer and written from a user's perspective. Unlike requirements, scenarios are not complete specifications but only examples. Scenarios do not have to be collected at the beginning of the project, but can be added and modified during development.

Thus, requirements are replaced by two artifacts: tests and scenarios.

1.3 Roles

Agile methods define various roles:

1. The *Team* is a self-organizing group of developers and other roles. Members of the team assign work items to themselves.
2. The *Product Owner* is responsible for defining the properties of the product under development. This encompasses the right to change those properties, but not while an iteration is ongoing.
3. The *Scrum Master* acts as a coach, mentor, guru, and method enforcer for the team. This role cannot be assumed by the same person that acts as the Product Owner.
4. The *Customer* is directly involved in the project or even a member of the team—rather than just being the source of requirements at the beginning and the recipient of the finished product at the end of the project.

1.4 Practices

The principles stated before are implemented using the following practices:

- Organizational

1. *Daily Meeting*: Every team member answers the following questions in a daily face-to-face meeting that takes no longer than 15 minutes: 1) “What did I do yesterday?” 2) “What do I plan to do today?” 3) “What impediments am I facing?” Those impediments are then resolved outside the daily meeting in order to keep it short.
2. *Planning Game*: Work items are estimated in a group setting, where participants are forced to come up with their own initial guess independently. Afterwards, a consensus is found iteratively in a group discussion.
3. *Continuous Integration*: Changes are integrated continuously (i.e. multiple times per day) rather than after longer development periods in order to detect incompatibilities early on.
4. *Retrospective*: The team reflects on the finished iteration in order to use the experience gained and lessons learned to improve the upcoming iteration.
5. *Shared Code Ownership*: The team as a whole, rather than individual programmers, are responsible for the entire code base. This is supposed to reduce the dependence on individuals and to avoid territorial conflicts.

- Technical

1. *Test-Driven Development*: A yet failing test case is written that describes the new functionality to be added, after which the actual functionality is implemented in order to make the new test case pass. The code can then be refactored as long as all the entire regression test suite passes.
2. *Refactoring*: The implementation is adjusted structurally in order to improve the design of the system. This step is crucial in conjunction with test-driven development to make sure that the code being written is not only working in terms of passing tests but also well-designed. Refactoring is the agile answer to big upfront design decisions of traditional methods.
3. *Pair Programming*: Code is developed by two programmers sharing a workstation in changing roles—the “driver” writing the code while explaining his thoughts, and the “navigator” trying to understand the driver’s thought process in order to catch mistakes early on and to provide criticism.
4. *Simplest Solution*: Software engineering principles to improve the code’s extensibility and reusability are ignored because—according to agile methods—the further direction of development cannot be known in advance. Instead, only the minimalistic product requested is built and delivered.
5. *Coding Standards*: Teams define style rules that are then applied to all the code being written.

1.5 Artifacts

Agile methods rely on a set of virtual and material artifacts:

- Virtual

1. *Use Case* and *User Story*: Both are scenarios describing an interaction of a user with a system. Use cases predate agile methods and cover an entire run through the system, whereas user stories originate from agile methods and only describe the interaction from the user's point of view.
 2. *Burndown Chart*: The amount of work items due for an iteration (y-axis) is plotted against time (x-axis) to measure the *velocity*: the amount of work done per unit of time. Since the amount of work items is fixed during an iteration, the plotted line is falling as tasks are finished (and not re-opened). The plotted line can quickly be compared against a planned linear velocity denoted by a straight falling line, which informs team members of their progress (and lack thereof) at a glance.
- **Material**
 1. *Story Card*: User stories shall be written down to small cards of the same size (3x5 inches in the US, DIN A7 in Europe), which therefore must be kept short.
 2. *Story Board*: The story cards for an iteration are pinned to a story board, where the card's location indicates the underlying user story's progress, for which columns titled "todo", "in progress", "testing", "done" or the like are used.
 3. *OpenRoom*: Development shall take in an open-plan setting as opposed to cubicles or closed offices, which favours interaction between team members instead of isolating them.

1.6 A First Assessment

Some of the ideas stated are genuine inventions of agile methods, while others are older and just have been re-discovered and popularized. Some of those ideas are good, others less so. Combining those qualities—new and old, good and bad—allows for clustering agile ideas into four categories:

- *not new, not good*: User stories and use cases document examples of interactions with a system. While they are useful to validate requirements, they are inadequate for replacing them. Without a proper requirements process, crucial engineering steps such as generalization and abstraction are omitted, resulting in poorly extensible and inflexible products.
- *new, not good*: While pair programming may be a useful technique in some cases, insisting on its constant use (as Extreme Programming does) ignores both different personalities of programmers and studies that fail to show its benefits. Furthermore, methods like Lean denounce useful engineering qualities such as generalization, extensibility, and automation as "waste", which might be appropriate to push back against unproductive perfection in some cases, but causes more harm than benefit when ignored altogether. Leaving out a systematic requirements process because requirements will change anyway is an inappropriate implication, because all kinds of artifacts in a software project are subject to change—but are being created nonetheless.

- *not new, good*: Iterative development with daily integration builds as well as early and frequent releases has been practiced both in commercial settings and in Open-source projects for decades. While agile methods insist on those practices, they have not invented them. The importance of embracing change is also not a new idea, but has been recognized long before. Ironically, some practices advertised by proponents of agile methods make change actually harder rather than facilitating it.
- *new, good*: Allowing a competent team to manage itself is certainly empowering. A daily meeting not only reinforces interaction between programmers, but also clarifies the next steps to be taken, and allows for early detection and resolution of impediments. Time-boxed iterations with frozen requirements during that period add stability to the development process without rendering it inflexible—change is only delayed for a couple of weeks and then handled properly. The focus on the regression test suite as a central artifact of development encourages both quality and professionalism. The focus on code as the project’s main deliverable—rather than diagrams and documents—is another idea that agile methods managed to convey to the software industry at large.

2 Deconstructing Agile Texts

Literature advocating agile methods of software development often resorts to intellectually unsound methods to convince its readers.

2.1 The Plight of the Traveling Seminarist

Consider the following example from Mike Cohn’s *Succeeding with Agile*, in which the author tries to convince the reader of the superiority of the spoken over the written word:

When the author asked his assistant to “book the Hyatt in Denver” by email, she responded with “the hotel is booked”, by which she meant that she was unable to book a room, because the hotel was already full. However, the author interpreted her response as “your room has been booked as requested”.

Cohn then argues that this misunderstanding could have been avoided had this exchange taken place by phone rather than by email. Therefore, he argues, discussions shall be favored over documents.

Ironically, this anecdote was originally presented orally in a seminar. It is way less convincing in the book’s written form—where the misunderstanding becomes evident to the reader immediately. Thus, this anecdote refutes its own argument.

It is also an overstretch to conclude that discussions are better suited for software development than written requirements: This is a mere *proof by anecdote*. An anecdote, however, cannot proof a generalization; it can only proof that a generalization does *not* hold by counterexample.

Also, an opposing anecdote of a misunderstanding that could have been avoided by requiring a written specification rather than just relying on the memory of a discussion would already be a sufficient counterargument to the initial “proof” by anecdote.

Usually, such arguments just continue by invoking more anecdotes, or even worse, quotes and aphorisms, but which nothing is ever concluded.

There are many reasons to write down requirements:

- The spoken word is more ambiguous than the written one. The issue of written requirements “looking more precise than they are” is better resolved by more formal notations rather than by resorting to the even more ambiguous spoken language.
- Many discussions end with the request to “write it down”, because a final decision often can only be made after reviewing a written request.
- Projects involving people from different cultures with their own English accents often suffer from misunderstandings in verbal discussions that become apparent immediately in the written form, after which the issue can usually be resolved quickly.
- Verbal discussions and their conclusions are only known to the people that attended them. A written version of the conclusions made can be shared with a wider circle.
- If decisions are made on the basis of verbal discussions, it is not certain whether or not the person making the decision is actually authorized to do so. A written document can be reviewed by the relevant stakeholders, unlike decisions being made by the last people that talked about an issue.
- People leave and join ongoing software projects. Unlike verbal discussions, written requirements survive such fluctuations.

Even though pyramids and cathedrals have been built based on mostly verbal interactions, modern engineering came a long way thanks to written requirements. Rejecting them because they are not always 100% precise is the wrong conclusion: more formal written requirements rather than fuzzy discussions are the way out.

Written words *can* be misleading: A written requirement for a system to respond “in real time” says nothing. However, here a more precise language such as mathematics ($t_{\text{answer}} \leq 0.1s$) rather than the more ambiguous verbal communication is the solution.

Communication *is* hard and can cause issues in every projects. Written documents sometimes are just not enough: it’s also important for the stakeholders to talk to one another. But this doesn’t mean that spoken language can replace the written word, quite the opposite: spoken language *complements* documents.

It’s true that written documents are time-consuming and expensive to create. But they can be archived and searched through—unlike verbal communication.

2.2 The Top Seven Rhetorical Traps

Advocates of agile methods often rely on rhetorical devices in their texts. Even though such unsound techniques do not disprove their argument, they should make the reader alert to attempts manipulation. The most common of those rhetorical devices are:

1. *Proof by Anecdote*: While anecdotes can be useful to illustrate a point, they can also backfire when a reader made different experiences than implied with the anecdote. Some anecdotes rely on analogies, which might invoke images in the reader's mind contrary to the author's intention. An anecdote is to a principle what a test is to a specification: it gives one example that cannot necessarily be extrapolated to the general case.
2. *Slander by Association*: Picking the word "agile", whose opposites all have negative meanings and connotations, is rhetorically very effective. Likewise, lumping all non-agile ideas together with things generally perceived as outdated and backwards ("Waterfall!") might be just as effective—but also dishonest.
3. *Intimidation*: Some ideas initially considered absurd turned out to be genius. However, this does not imply that *any* idea sounding absurd is equally valid. This logical fallacy follows the form "B implies A from A implies B" or "*anything* with quality X is good, because *something* with quality X is good". Any expert's criticism is rejected based on anecdotes in which experts have been proven wrong. Using the establishment as the enemy can be an effective way to rally others behind a revolutionary idea. And if evidence contrary to such an idea shows up, simply the environment, in which the idea has been tried, can be blamed: If agile does not work in your organization, your organization is the problem. Anybody not buying into the agile way of handling projects is labeled as an authoritarian member of the command and control gang.
4. *Catastrophism*: Advocates of a new method need to point out the flaws of existing ones. The term "software crisis" became prevalent in 1968 and has been used ever since to paint a bleak picture of software development and its results. However, a lot of software projects are successful, and the software created thereby performs its duty observably well. Not everything is as bad with the current state of software and the—"pre-agile"—process that created it as proponents of agile methods often claim.
5. *All or Nothing*: It is fair for a methodology to define a set of principles its users have to adhere to in order to make it work. However, such a set of absolute requirements must be kept small for a methodology to be practicable. Blaming failed projects for not totally sticking to the prescriptions while celebrating successful projects cherry picking just a couple of such practices of a methodology is dishonest. One recipe does not fit every organization and project; principles and practices have to be selected as it fits.
6. *Cover Your Behind*: Many agile texts advocate extreme ideas and back those up with some words of warning where those ideas will not necessarily work—but fail to provide useful criteria for such a delimitation. Such tempering serves the authors well for covering their backs ("I didn't tell you to *not* supervise the team/gather requirements/plan the project!"), but leaves the reader without any guidance whether or not he shall implement a particular idea in his specific circumstances. This rhetorical device is often stated as "X does not mean Y" when X and Y are hardly distinguishable. One example from *Lean*

claims that “deciding as late as possible does not mean procrastinating”—where “deciding as late as possible” is the literal dictionary definition of “procrastinating”.

7. *Unverifiable Claims*: Many agile claims—“doing twice the work in half the time”, i.e. improving productivity by a factor of four—are simply too good to be true. While it’s possible that a formerly demotivated team run by incompetent managers can become hugely productive over night when given the chance to work on their own accord, such examples cannot be generalized to all circumstances. Software projects are not repeatable lab experiments: they are influenced by their surrounding environments and people working at them as well as by their experiences made in the process.

3 The Enemy: Big Upfront Anything

Agile methods reject anything “plan-based” or “predictive”. Such approaches are lumped together with the term “waterfall” to denounce it using slander by association, thereby ignoring that the original 1970 paper used the Waterfall Model as a strawman argument how software development does *not* work. This view also ignores that engineering is by definition predictive to some extent, using methods of science and management to some degree.

3.1 Requirements Engineering

Software engineering is not just programming. It also involves *requirements analysis*: figuring out the problem a stakeholder wants to solve and appropriate solutions to it. While most of software engineering is about building the system right, requirements are about building the right system.

A big part of requirements engineering is *requirements elicitation*: gathering user needs. This encompasses, among others, stakeholder interviews (asking people what they need) and workshops (discussing and clarifying requirements to identify and resolve conflicts).

The results of this process are usually gathered in a *requirements document*. Proponents of agile methods criticize both process and result using two different arguments:

1. *Waste*: A requirements document is not a useful deliverable for the customer. Some gathered requirements will not be implemented, and gathering them is therefore wasteful.
2. *Change*: Customers do not know what they want and will change their minds as they get a chance of interacting with the system. Requirements become obsolete; it is better to create new versions of the software iteratively based on user feedback than to lock down their initial ideas.

There are two approaches to reduce waste: The plan-first approach reduces waste by first gathering and prioritizing requirements and then dropping the ones considered not essential. The agile approach reduces waste by producing a minimal solution, of which parts are dropped the customer deems unnecessary.

While the first approach produces one sort of waste—unused requirements—it avoids another form of waste—unused implementations, which are common in agile projects—and can be very frustrating for developers. Both techniques have the same goal and can be used in conjunction.

Large and detailed requirements documents are overkill for many projects. This does not imply that no requirements shall be gathered whatsoever, but that the level of detail needs to be adjusted to fit the project. And even though a requirements document is not a deliverable, it serves as the foundation of a *system documentation*, which is delivered to the customer.

The argument concerning change is correct in so far as requirements cannot be frozen after initially gathering them. However, having gathered an initial set of requirements does not imply carving them in stone. Requirements are just one of many artifacts of a software project alongside code and regression tests—which change all the time and are written nonetheless! Just because plans need to be adjusted does not mean that plans are useless.

Furthermore, an initial requirements *phase* does not imply that the resulting requirements *document* must remain static. Requirements can be put under version control along the code that implements them.

A proper requirements process distinguishes between two kinds of requirements describing different aspects:

1. *Domain*: properties of a model as part of the world in which the system will operate
2. *Machine*: desired properties of the system the project wants to build

As agile methods see requirements as design or candidate solutions, they fail to make this important distinction: It is *not* up to the project to define the domain, which is given as business rules, regulations, laws, or properties of the physical world. Such restrictions are *gathered* rather than made up. The project is free, however, to define the machine requirements—as design decisions, so to speak.

The agile critique is right that often too much time is spent on figuring out design and implementation details to be disguised as mere requirements. It's often better to wait until more reliable information becomes available.

3.2 Architecture and Design

In software, the code expresses the actual solution. The *design* of a software defines its *architecture*: its modular structure, the abstractions chosen, the design patterns used, and the interfaces specified.

It has long been recognized that in software development, there is no sharp distinction between design and implementation, because the implementation has major impacts on the properties of the system as a whole, e.g. on its performance.

Compared to an industrial process, the compilation rather than the writing of the source code comes closest to the production phase. Therefore, the word “design” has a different meaning in software development than in, say, mechanical or electrical engineering.

There is also no firm line between code and documentation, given that code contains comments, and artifacts such as UML diagrams resemble code quite closely to be useful.

Code always has an architecture—as a result of a distinct design phase or as it emerged from development. Just like a mathematical paper that looks much orderlier than the process that created it, a software design can be the result of a more or less structured approach.

While agile methods denounce a separate upfront design phase, they have no common approach to design, but some guiding principles:

1. Design in iterations alternating with regular implementation iterations rather than in a separate upfront design phase.
2. Solve the problem at hand rather than focussing on extensible and reusable design.
3. Create a working solution and refactor it rather than aiming at a perfect initial solution.

The agile literature admits that approach makes planning the work and assigning it to team members much harder. Not having a big architecture picture can also make the work on the system under development harder. Reworking its parts becomes inevitable, too.

This view on design is sensible insofar as it is not a good idea to do too much design at the beginning of a project. However, the categorical rejection of any upfront design activity also has its issues:

1. Refactoring only works well in a code base with a decent design to begin with. This is hardly the case if no thoughts are spent on extensibility and reusability at all.
2. Some design decisions—or the failure to take them—are extremely costly to revert, e.g. the decision between a monolingual and a multilingual user interface.

Just like security, good design requires both some upfront conceptual work and an ongoing effort. Here, the total agile rejection of any upfront design process goes too far.